

Series and infinite products – 7

Infinite products

12 сентября 2026 года

Theoretical minimum.

Definition and Theorem 1 (criterion of convergence). The product $\prod(1 + a_n)$ is called convergent if the partial products tend to a *non-zero* limit. For $a_n > -1$ this happens exactly when $\sum \ln(1 + a_n)$ converges, and the product converges absolutely if and only if $\sum |a_n| < \infty$. If $\sum a_n$ converges while $\sum a_n^2$ diverges, the product diverges to zero: the expansion $\ln(1 + a) = a - \frac{a^2}{2} + O(a^3)$ shows what exactly must be controlled.

Theorem 2 (telescoping products). If $1 + a_n = \frac{b_{n+1}}{b_n}$, the product collapses to $\lim \frac{b_{N+1}}{b_1}$. A rational $1 + a_n$ is telescoping as soon as its numerator and denominator factor into linear factors whose roots differ by integers, and the answer is then a ratio of gamma values.

Theorem 3 (the three classical products).

$$\sin \pi x = \pi x \prod_{n \geq 1} \left(1 - \frac{x^2}{n^2}\right), \quad \frac{\pi}{2} = \prod_{n \geq 1} \frac{4n^2}{4n^2 - 1}, \quad \frac{\sin \theta}{\theta} = \prod_{n \geq 1} \cos \frac{\theta}{2^n}.$$

The first is Euler's product, the second is Wallis' formula (obtained from it at $x = \frac{1}{2}$), and the third at $\theta = \frac{\pi}{2}$ is Viète's product — the doubling map again, as in the supplement on trigonometric and hyperbolic substitutions.

Fact 1 (products into sums). Taking the logarithm is the default move, but it is not always the cheapest: a product of the form $\prod(1 - \frac{1}{a_k})$ with a fast recurrence for a_k is usually telescoping after the substitution $a_k = z_k + z_k^{-1}$, and a product over an arithmetic family is often the Euler product of a Dirichlet series (Series and infinite products – 6).

1. Prove Theorem 1 in full, and show all three phenomena on examples: a product with $\sum a_n$ divergent but the product convergent; a product with $\sum a_n$ convergent which diverges to zero; and the behaviour of $\prod(1 + \frac{(-1)^n}{\sqrt{n}})$.
2. (Putnam 1977) Evaluate the infinite product $\prod_{n=2}^{\infty} \frac{n^3-1}{n^3+1}$.
3. (OMOUS 2023) Compute $\lim_{n \rightarrow \infty} \sqrt[n]{\prod_{k=1}^n \Gamma(\frac{k}{n})}$, where Γ is the gamma function. (The telescoping product $\prod_{n \geq 3} \frac{(n^3+3n)^2}{n^6-64}$ of IMC 2019 is the model example of Theorem 2 and has already been done in Limits – 1.)
4. (Putnam 2014) Let $a_0 = \frac{5}{2}$ and $a_k = a_{k-1}^2 - 2$ for $k \geq 1$. Compute $\prod_{k=0}^{\infty} \left(1 - \frac{1}{a_k}\right)$ in closed form.
5. (AKHIMO 2025) The function sinc is defined on $\mathbb{R}$ by $\text{sinc}(x) = \frac{\sin x}{x}$ for $x \neq 0$ and $\text{sinc}(0) = 1$. Prove that

$$\prod_{n=1}^{\infty} \text{sinc}\left(\frac{x}{2n-1}\right) = \prod_{n=1}^{\infty} \cos\left(\frac{x}{2n}\right) \quad \text{for all } x \in \mathbb{R}.$$